

VARUN MANDEPUDI

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EDUCATION

Master's in Data Science

Sept 2025 – Present

Northeastern University, Boston, Massachusetts, USA; GPA: 3.000

Expected Graduation: May 2027

Related Courses: Algorithms, Supervised Machine Learning, Data Processing and Data Management.

B. Tech Computer Science Engineering with Specialization in Big Data Analytics

Sept 2021 – May 2025

SRM University, Chennai, Tamil Nadu, India; CGPA: 9.32

Related Courses: Data Structures and Algorithms, Operating Systems, Discrete Mathematics, Machine Learning.

TECHNICAL KNOWLEDGE

Programming & Tools: Python (Pandas, NumPy, Scikit-Learn, TensorFlow), R, SQL, Jupyter, Excel, Power BI, Tableau.

Machine Learning: Regression, Classification, Clustering, Time-Series Forecasting, Model Evaluation.

Data Engineering: Feature Engineering, Data Cleaning, RAG Pipelines, Data Warehousing

PROFESSIONAL EXPERIENCE

Data Analyst Intern – Thaaai Casting Limited

Sept 2024 – Nov 2024

- Analyzed company sales data using clustering and classification techniques.
- Identified patterns in capital fluctuations and underperforming regions.
- Analyzing production logs from Pressure Die Casting Machines.
- Developed and presented Power BI visualization reports.
- Recommended ideas that led to an 8% increase in regional sales.

ACADEMIC PROJECTS

RAG Chatbot for Computer Science Research Papers

Sep 2025 – Dec 2025

Objective: Build a Retrieval-Augmented Generation (RAG) system to answer questions on computer science research papers using arXiv data.

Role: Designed and implemented the database and data pipeline using PostgreSQL and FAISS, managed large-scale metadata and embeddings (547k+ papers), developed SQL analytics for evaluation, and integrated semantic search with a local LLM (Mistral via Ollama) to generate source-grounded responses through a Flask web interface.

Energy Consumption Forecasting

Aug 2024 – Oct 2024

Objective: Develop an ensemble model incorporating machine learning to predict energy consumption

Role: Collected data and developed an ensemble incorporating three machine learning models, namely Random Forest regressor, XG Boost Regressor, and Long Short Term Memory Network (LSTM), and evaluated the accuracy

Sentimental Analysis on Social Media Data

Apr 2024 – May 2024

Objective: Analyze sentiments in social media posts to understand public opinion

Role: Collected data and classified sentiments as positive, negative, or neutral; Applied NLP techniques and machine learning models for sentiment classification; Visualized and displayed sentiment trends and distributions

PUBLICATIONS

Submitted the research paper on **Waste Classification using Machine Learning Models: A Comparative Study** to the International Conference on Research and Development in Information, Communication and Computing Technologies (ICRDICCT'25) - Accepted and Presented at the Conference on Apr 4 & 5, 2025. (Awaiting Publishing)

- Performed comparative analysis of various machine learning models to identify the most effective approach for automated waste classification.

Submitted the research paper on **Transformative Potential of Artificial Intelligence and Machine Learning in the Energy Sector: Smart Grids** to the International Conference on Progressive Computational Intelligence, Information Technology and Networking (Com-IT-Con, 2024) – Presented and Published at the Conference on Oct 25, 2024

- Developed an ensemble model incorporating three machine learning models: Random Forest regressor, XG Boost Regressor, and Long Short-Term Memory Network (LSTM) for predicting future energy consumption